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Question 1

Not answered

Marked out of 1

A CPD programme for mental health practitioners is aimed at improving communication skills. A quiz is administered prior to the start of the programme and again following the completion. The practitioners attending the programme are at the 4th percentile of quiz score at pretest. At posttest, they are around the 20th percentile. Apart from concluding that the programme is very effective which of the following could explain the change in scores

Select one:

- ☐ Selection bias
- ☐ Confounding bias
- ☐ Repeat testing effect
- ☐ Low sample size
- ☐ Measurement error

Your answer is incorrect.

Repeatedly measuring the participants may lead to bias. Participants may remember the correct answers or may be conditioned to know that they are being tested. Repeatedly taking (the same or similar) verbal performance tests usually leads to scoring gains - beware of being falsely reassured when taking the same mock exam before and after a one-day course!

The correct answer is: Repeat testing effect

Question 2

Not answered

Marked out of 1

Patients in the control groups of randomized clinical trials seem to have a better clinical course than similar patients who do not participate in trials. All of the following may contribute to this trial participation effect except

Select one:

- ☐ Use of higher potency of medications
- ☐ Selection criteria for clinical trials may screen out the sickest patients
- ☐ Frequent monitoring and extra clinical care
- ☐ Placebo effect
- ☐ More experienced clinicians and academic centres

Your answer is incorrect.

It is more likely that participation in clinical trials by itself (irrespective of the intervention received) has a positive effect on the outcome of patients. The suggested reasons include better clinical input with experienced research-clinicians, better treatments, favourable selection criteria, the placebo effect (or participation effect) and better use of treatment protocols. BMJ 2005;330:1164 (21 May)

The correct answer is: Use of higher potency of medications

Question 3

Not answered

Marked out of 1

Which one of the following would not be expected to increase the statistical power of a clinical trial to test a new medical treatment

Select one:

- ☐ Lengthening the trial to allow enrollment of more patients
- ☐ Adopting stronger measures to ensure compliance with therapy
- ☐ Treating a more clinically homogeneous group of patients
- ☐ Adopting a more stringent standard of evidence to prove the new treatment works
- ☐ Testing only a subgroup of patients for whom the new drug would be expected to have the greatest advantage

Your answer is incorrect.

The general formula for calculating a test statistic is $\text{test statistic} = (x - \mu) / (s / \sqrt{n})$ where x = the sample estimate, μ = the hypothesized value, and $s / \sqrt{n}$ the standard error of x . Power (ability to detect true difference if it exists) would be affected by the effect size being tested ($x - \mu$), the sample size (n), and the variability (standard deviation) among the samples (s). Adopting a more stringent standard of evidence means it will be difficult to demonstrate significance in smaller samples - one will require larger sample size to generate the required power. Enrolling more patients, testing a subgroup with a higher probability of hypothesised outcome or a more homogeneous group (i.e. lower variability) and ensuring adherence (low dropouts, low missing data), etc. can increase the power.

The correct answer is: Adopting a more stringent standard of evidence to prove the new treatment works

Question 4

Not answered

Marked out of 1

A drug firm reanalysed its data submitted for licensing an antidepressant ten years ago. The analysis indicated that the Number needed to harm NNH for the suicidal act when using the antidepressant was 60. The original data included 500 subjects in placebo arm out of which 50 exhibited suicidal acts. There were 600 subjects in the active antidepressant arm. Calculate the number of subjects in the active arm that exhibited suicidal acts

Select one:

- ☐ 3
- ☐ 70
- ☐ 60
- ☐ 5
- ☐ 8

Your answer is incorrect.

Calculating NNH needs calculation of ARR (Absolute risk reduction). As NNH for antidepressants is 60, this means the event rate must be higher for antidepressants than placebo. In other words, the risk of a side effect is likely to be lower with placebo.

$ARR = \text{experimental event rate} - \text{control event rate}$ (i.e., $EER - CER$)

$NNH = 1/ARR$. $ARR = 1/NNH = 1/60$ for this study.

Control event rate is given as 50/500. We are asked to calculate no. of experimental events. This can be derived from experimental event rate.

So here $ARR = (EER - (50/500))$.

Replacing known values, $1/60 = ((\text{no. of experimental events}/600) - (1/10))$

$\text{No. of experimental events}/600 = (1/60) + (1/10)$

$\text{No. of experimental events}/600 = (1/60) + (6/60) = (7/60)$

No. of experimental events = $(1/60) + (6/60) = (7/60) * 600 = 7 * 10 = 70$

The correct answer is: 70

Question **5**

Not answered

Marked out of 1

In a study comparing drug A and placebo control, 20 out of 200 patients taking drug A die after three years. 25 out of 225 patients taking the placebo die after three years. If death is the outcome of interest, the number of patients who need to take drug A to prevent one death is

Select one:

- ☐ 9000
- ☐ 10
- ☐ 1
- ☐ 90
- ☐ 9

Your answer is incorrect.

In this case $ARR = 1/90$. Hence, NNT will be 90. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 82

The correct answer is: 90

Question 6

Not answered

Marked out of 1

Which of the following statements regarding non-adherence and intention to treat analysis in RCTs is true

Select one:

- ☐ Intention-to-treat analysis distorts the principle of randomisation
- ☐ The prognostic factors are comparable between adherent and non-adherent subjects
- ☐ Subjects randomised to placebo arm but did not adhere to prescription have better prognosis than those who adhere to placebo.
- ☐ A subject who dies before receiving the first dose of placebo must be excluded from intention-to-treat analysis
- ☐ A subject who gets 'cured' before completing the trial must be included in intention-to-treat analysis

Your answer is incorrect.

Removing patients after randomization always risks introducing bias by creating noncomparable groups. Hence, Intention to treat analyses must include those who are randomised but not received the intended intervention in the final analysis. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 173

The correct answer is: A subject who gets 'cured' before completing the trial must be included in intention-to-treat analysis

Question 7

Not answered

Marked out of 1

A study reports that 20% of a control group had a relapse of schizophrenia during the trial period, while only 15% of the treatment group relapsed. Which of the following is a valid statement based on the above information

Select one:

- ☐ The treatment is 15% effective in reducing the relapse
- ☐ The outcome has been treated as a dichotomous variable
- ☐ The relative risk of relapse in treatment group compared to controls will be >1
- ☐ The treatment can prevent 20% of relapses
- ☐ The treatment reduces the risk of death by 5%

Your answer is incorrect.

As the results are in proportion, one can conclude safely that the variable was dichotomous. The numbers 15% and 20% are event rates - they are not risk reduction rates. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 1st ed. McGrawHill 2002 Pg66

The correct answer is: The outcome has been treated as a dichotomous variable

Question 8

Not answered

Marked out of 1

To be statistically significant, the confidence intervals of relative risk reduction achieved by an intervention should not cross

Select one:

- ☐ Zero
- ☐ 100
- ☐ Infinity
- ☐ 10
- ☐ One

Your answer is incorrect.

The value for a statistic corresponding to no effect in the population is called the null value. For correlations, risk difference, relative risk reduction (RRR) and changes in the mean, the null value is zero. If the outcome statistic is a relative risk or odds ratio, the null value is 1 (equal risk or odds). So for RRR the result is statistically significant if the confidence interval does not overlap zero.

The correct answer is: Zero

Question 9

Not answered

Marked out of 1

An article reporting a trial does not provide confidence intervals around the point estimate of relative risk. If the p-value achieved by the treatment effect is exactly $p=0.05$, then the lower bound of the 95% confidence limit for the relative risk is likely to be

Select one:

- ☐ 100
- ☐ 10
- ☐ Zero
- ☐ Infinity
- ☐ One

Your answer is incorrect.

For a statistically non-significant result (i.e. p-value more than 0.05) the lower confidence limit of RR might have been less than 1. As the p-value is exactly 0.05, the lower confidence limit of RR is likely to be one.

The correct answer is: One



Question 10

Not answered

Marked out of 1

95% CI for the relative risk reduction achieved by an intervention in an expensive multicentre RCT is -28% to 49%; Which of the following conclusions can be valid

Select one:

- ☐ 28% of patients will show a negative effect
- ☐ More people who take the intervention will have positive response than negative response
- ☐ The RCT really has not helped to decide whether or not to offer the new treatment
- ☐ 95% of patients will have treatment response between -28% and 49%
- ☐ The maximum benefit that can be achieved by the drug is 49%.

Your answer is incorrect.

These results are consistent with both an RRR of -28% (that is, patients given the new treatment might be 28% more likely to die than control patients) and an RRR of nearly 49% (that is, patients subsequently receiving the new treatment might have a risk of dying almost 50% less than those who are not treated). In other words, the 95% CI on this RRR is -28% to 49%, and the trial really has not helped us decide whether or not to offer the new treatment. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008 Pg 77

The correct answer is: The RCT really has not helped to decide whether or not to offer the new treatment

Question 11

Not answered

Marked out of 1

A trial randomized 100 patients each to treatment and control groups; there were 20 relapses in the control group and 15 relapses in the treatment group. What will happen to the point estimate of relative risk reduction if the sample size is increased to 1000 a group, but the event rates remain stable

Select one:

- ☐ The point estimate is more likely to cross the zero value
- ☐ The 95% confidence intervals around the estimate become less precise
- ☐ The point estimate moves farther away from the true population value
- ☐ Point estimate of RRR stays the same
- ☐ The upper bound of the 95% limits is most likely to be in negative values

Your answer is incorrect.

As RR is EER/CER, it will not change by merely increasing sample size but having same event rate (though this may be nearly impossible practically). As RR does not change, RR (which is $1 - RR$) may not change. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008Pg78

The correct answer is: Point estimate of RRR stays the same

Question 12

Not answered

Marked out of 1

A group therapist leading a smoking cessation group becomes curious about the effectiveness of new drug varenicline on smoking cessation. She approaches an interested pharmacist, and together they recruit 60 smokers attending the group. She asks for volunteers to undergo the drug trial. 30 volunteers are given varenicline for 12 weeks. At the end of 12 weeks 'frequency of smoking urges' is compared using a validated scale. Which of the following components of a drug evaluation trial is missing in this experiment

Select one:

- ☐ Informed consent
- ☐ Proper outcome measure
- ☐ Clear hypothesis
- ☐ Randomisation
- ☐ Intervention

Your answer is incorrect.

Lack of randomisation is a serious threat to the external validity of intervention trials. A sample may have certain features that may interact with the independent variable, limiting generalizability. Note that the inferences here are based on volunteers in a smoking cessation group. It is likely that non-volunteers have a poor motivation and may respond in a significantly different fashion to what is demonstrated by this study - randomised sampling would have helped in generalisation of the results.

The correct answer is: Randomisation

Question 13

Not answered

Marked out of 1

Which of the following conditions encourage acceptance of a result from subgroup analysis in an RCT

Select one:

- ☐ Primary hypothesis is not proven but subgroup analysis shows a significant result
- ☐ Multiple subgroups are analysed but only one showed a clear statistical significance
- ☐ The subgroup hypothesis was specified before the study began
- ☐ Other studies fail to replicate the subgroup findings
- ☐ The effect size for subgroup falls within the 95% confidence limits of the primary outcome

Your answer is incorrect.

Most proponents of EBM advise us to remain skeptical of subgroup analyses. A treatment is likely to benefit the subgroup more or less than the other patients only if the difference in the effects of treatment in the subgroups is large and very unlikely to occur by chance. Even when these conditions apply, the results may be misleading if investigators did not specify their hypotheses before the study began, if they had a very large number of hypotheses, or if other studies fail to replicate the finding. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008 Pg80

The correct answer is: The subgroup hypothesis was specified before the study began

Question 14

Not answered

Marked out of 1

Which of the following is NOT a surrogate or substitute endpoint for a treatment trial

Select one:

- ☐ Working memory scores in donepezil trials for schizophrenia patients with negative syndrome
- ☐ Impulsivity score in atomoxetine trials for ADHD
- ☐ Activities of daily living scores in dementia trials
- ☐ Lipid levels in a trial of statins for antipsychotic-related metabolic syndrome
- ☐ Glycosylated haemoglobin levels when using rosiglitazone for patients with clozapine induced diabetes.

Your answer is incorrect.

Surrogate outcomes are often attractive for reducing sample size and decreasing the length of follow-up and sometimes even enabling ethical approval of studies. Unfortunately, they can be misleading. Any outcome that is easily measurable but used as a proxy for the actual primary outcome of direct benefit to patients is a surrogate outcome. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008 Pg81

The correct answer is: Activities of daily living scores in dementia trials

Question 15

Not answered

Marked out of 1

In a patient with many risk factors for metabolic syndrome and taking risperidone for schizophrenia, the 3-year risk of stroke is estimated to be 30%. If he takes co-prescribed statins during this period, this risk drops to 20%. What is the NNT to prevent stroke in this patient

Select one:

- ☐ 1
- ☐ 5
- ☐ 4
- ☐ 3
- ☐ 10

Your answer is incorrect.

ARR = 10%. Hence NNT = $100/10 = 10$.

The correct answer is: 10



Question 16

Not answered

Marked out of 1

Which of the following is true concerning the interpretation of a randomised trial results

Select one:

- ☐ Relative risk reduction is the same as risk difference
- ☐ Relative risk is obtained by subtracting 1 from odds ratio
- ☐ Odds ratio must never be equated to relative risk in RCTs
- ☐ Relative risk is generally far higher than risk difference (ARR)
- ☐ If the event rates are very high odds ratio may be equivalent to relative risk

Your answer is incorrect.

Failing to distinguish between the OR and the RR when interpreting randomized trial results will seldom mislead us. However, we must correctly distinguish between the RR and the RD (risk difference or ARR). This is because the RR is generally far larger than the RD, and presentations of results in the form of RR (or RRR) can convey a misleading message. For example, reducing a patient's risk by 50% sounds impressive. That may, however, represent a reduction in risk from 2% to 1%. The corresponding 1% ARR sounds considerably less impressive. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg90

The correct answer is: Relative risk is generally far higher than risk difference (ARR)

Question 17

Not answered

Marked out of 1

Which of the following is correct concerning NNT to prevent an event of relapse in a treatment trial

Select one:

- ☐ NNT is inversely proportional to the baseline relapse rate
- ☐ NNT is directly proportional to the relative risk reduction
- ☐ For a given population, a more effective treatment with twice the RRR will double the NNT
- ☐ If the risk of relapse in your clinical population is two times higher, you have to treat twice the number of patients to prevent one relapse.
- ☐ If the same treatment is applied to a different population with half the relapse rate, the NNT halves.

Your answer is incorrect.

Assuming a constant RRR, the NNT is inversely related to the proportion of patients in the control group who have an adverse event (CER). If the risk of an adverse event doubles (for example, if we deal with patients at a higher risk of depression than those included in the clinical trial of antidepressants), we need to treat only half as many patients to prevent an adverse event; if the risk decreases by a factor of 4 (patients are younger, have less comorbidity than those in the study), we will have to treat 4 times as many people. The NNT is also inversely related to the RRR. With the same baseline risk, a more effective treatment with twice the RRR will reduce the NNT by half. If the RRR with one treatment is only a quarter of that achieved by an alternative strategy, the NNT will be four times greater. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg90

The correct answer is: NNT is inversely proportional to the baseline relapse rate

Question 18

Not answered

Marked out of 1

An investigator is interested not only in the total number of new-onset hyperlipidaemia following antipsychotic use, but also in the timing of onset after initiating the treatment. So he attempts to calculate an average measure of the relative risk weighted for number of patients at risk over a long period of follow-up in the trial cohort. This measure is best described as

Select one:

- ☐ Effect size
- ☐ Odds ratio
- ☐ Hazard ratio
- ☐ Standardised mean difference
- ☐ Degree of freedom

Your answer is incorrect.

The weighted RR over the entire study in a survival analysis is known as the hazard ratio.

The correct answer is: Hazard ratio



Question 19

Not answered

Marked out of 1

Which of the following cannot be identified from confidence intervals of an outcome measure such as relative risk in a treatment trial

Select one:

- ☐ The range within which the true treatment effect is likely to lie
- ☐ The degree of certainty attached to a range of values within which the true effect lies
- ☐ The statistical significance of a trial result
- ☐ The precision of a trial result
- ☐ The proportion of patients in the population who will have the estimated value of outcome

Your answer is incorrect.

The range within which the effect is likely to lie can be given by confidence intervals of a point estimate. Precision and degree of certainty can be given by confidence limits. One can understand statistical significance using p-values. The proportion of patients in the population who will have the estimated value of outcome cannot be determined from a trial estimate of RR

The correct answer is: The proportion of patients in the population who will have the estimated value of outcome

Question 20

Not answered

Marked out of 1

Which of the following is an appropriate way of interpreting negative trial results?

Select one:

- ☐ NNT is called NNH if the point estimate fails to reach preset level of statistical significance
- ☐ If the upper bound of confidence intervals contain an important treatment benefit, a beneficial effect cannot be ruled out
- ☐ If the confidence intervals cross point of no effect, the treatment is conclusively useless
- ☐ If the p value of point estimate is >0.05 , the treatment is harmful
- ☐ If the p value of point estimate is >0.05 , the treatment offers no benefit

Your answer is incorrect.

An apparently negative trial generally has a P value greater than 0.05; so it fails to exclude the null hypothesis that treatment and control interventions do not differ). In this case, one can focus on the upper bound of the confidence interval (that is, the end that suggests the largest benefit from treatment). If this value excludes any important benefit of treatment, we can conclude that the trial is definitively negative. But if the confidence interval includes an important benefit, the possibility that the treatment still might be worthwhile must not be ruled out. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 104

The correct answer is: If the upper bound of confidence intervals contain an important treatment benefit, a beneficial effect cannot be ruled out

Question **21**

Not answered

Marked out of 1

Which of the following comments regarding validity and reliability of research studies is correct

Select one:

- ☐ Random errors affect validity more than reliability
- ☐ Biased studies produce valid results which are unreliable
- ☐ Smaller samples yield results with less validity
- ☐ Large sample size reduces the likelihood of bias
- ☐ Random errors have more influence on the results of smaller studies

Your answer is incorrect.

In a study with large sample size, random errors become less prominent while in studies of smaller sample size the likelihood of a misleading result through random error is high. But note that bias (a threat to validity) does not disappear by merely increasing sample size. Biased studies can produce results that can be reproduced, but still not true. While random errors affect reliability more, bias affects validity to a higher degree. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 112

The correct answer is: Random errors have more influence on the results of smaller studies

Question 22

Not answered

Marked out of 1

Choose one correct statement with respect to randomised trials that are stopped early

Select one:

- ☐ Perceived harm due to the intervention does not warrant early cessation
- ☐ RCTs that are stopped early yield more data on safety and toxicity
- ☐ Early cessation is unethical under all circumstances
- ☐ The most common reason for premature cessation is early appearance of a large treatment effect
- ☐ Early cessation of RCTs generally underestimates the treatment effect

Your answer is incorrect.

Investigators may stop randomized controlled trials (RCTs) earlier than planned because of perceived harm of the experimental intervention, because they lose hope in achieving a positive result, or because the sponsor wishes to save money. The most common reason for early stopping, however, is that investigators note treatment effects that appear to be unlikely by chance (and that are often large) and that persuade them that the experimental intervention is beneficial. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 161

The correct answer is: The most common reason for premature cessation is early appearance of a large treatment effect

Question **23**

Not answered

Marked out of 1

Which of the following types of RCT is not subject to confounding by age, sex, or other individual variables obviating the need for examination of distribution of baseline characteristics following randomisation

Select one:

- ☐ Pragmatic trial
- ☐ Effectiveness trial
- ☐ RCT with three arms of comparison
- ☐ Cross over trial
- ☐ Factorial RCT

Your answer is incorrect.

Crossover clinical trials are not subject to confounding by age, sex, or most personal characteristics. But they are not useful in studies using irreversible endpoints such as 'complete cure' from a disease or death event. They need fewer people than parallel group trials as each individual plays the part of a 'trial subject' twice.

The correct answer is: Cross over trial

Question **24**

Not answered

Marked out of 1

In which of the following circumstances is an n-of-1 trial most suitable

Select one:

- ☐ To compare the effectiveness of two different interventions for a self limiting condition and open cholecystectomy in female patients
- ☐ Presence of RCTs demonstrating some beneficial effect for the intervention in study samples
- ☐ To evaluate treatment response for a patient with an acute and rapidly evolving illness
- ☐ To evaluate the effectiveness of a medication in preventing stroke
- ☐ To compare laparoscopic and open cholecystectomy in female patients

Your answer is incorrect.

N of 1 trials are most useful when one or several RCTs may have shown that the treatment in question is highly effective but 50% or more of patients in such trials have proved unresponsive, leaving the question whether our patient will respond or not. No useful result can be achieved if one attempts to experiment with a completely unknown intervention in n-of-1 fashion. For self-limiting conditions, preventive interventions, in rapidly changing illnesses and for irreversible treatments, n of 1 model may not be suitable. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 183

The correct answer is: Presence of RCTs demonstrating some beneficial effect for the intervention in study samples

Question 25

Not answered

Marked out of 1

In which of the following circumstances an n-of-1 trial is not recommended

Select one:

- ☐ The patient and clinician differ in their opinion regarding the usefulness of a certain drug which the patients insists to prescribe
- ☐ To find the optimal dose of a medication the patient should receive.
- ☐ To evaluate a treatment that has potentially curative effect on a longstanding condition though evidence is weak
- ☐ The patient has been started on a medication that has inconclusive evidence with regard to beneficial effect
- ☐ A patient has symptoms that are suspected to be an adverse effect

Your answer is incorrect.

There are many situations in which one can resort to n of 1 trials. 1. When a clinician is uncertain whether a treatment that has not yet been started will work in a particular patient. 2. The patient has started taking a medication, but neither patient nor clinician is confident that a treatment is really providing benefit. 3. Neither the clinician nor the patient is confident of the optimal dose of a medication the patient is receiving or should receive. 4. A patient has symptoms that both the clinician and the patient suspect—but are not certain—are caused by the adverse effects of the medications. 5. The patient has so far insisted on taking a treatment that the clinician believes is useless or harmful, and although logically constructed arguments have not persuaded a patient, a negative result of an n-of-1 RCT might. But if a condition disappears on treatment fully within the period of observation, it may not be possible to conduct n of 1 trials. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 183

The correct answer is: To evaluate a treatment that has potentially curative effect on a longstanding condition though evidence is weak

Question 26

Not answered

Marked out of 1

A trial was designed to evaluate the effect of treating post-MI depression using SSRIs or CBT on cardiac events. The primary outcome studied was a composite measure of the occurrence of death, recurrent MI or cardiac failure. All of the following justify using such a composite endpoint except

Select one:

- ☐ Follow-up period in the trial can be shortened
- ☐ The treatment will act more or less the same way across various components of the composite measure
- ☐ Need for huge sample size can be avoided
- ☐ There is uncertainty about which single end point to choose
- ☐ Patients consider some component endpoints of the composite measure to be more important than the others

Your answer is incorrect.

Composite end points (CEPs) capture the number of patients experiencing any one of several adverse events— e.g. death, MI, or hospitalization—as a primary study endpoint. The main reason for employing CEPs is to decrease the necessary sample size and duration of follow-up. This may be justified especially when there is a paucity of events in any single category and a presumption that treatment will act in more or less the same way across categories of adverse events. We can also use CEPs when there is an uncertainty about the single best outcome measure. But patients almost invariably assign varying importance to different health outcomes (Do you think patients will consider death, recurrent MI and failure in same terms?). Hence, we must not ignore the possible difference of treatment effect on individual component endpoints on the grounds that they share identical patient-importance. Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 242 and van Leth, F. and Lange, J. M. A. (2003) Use of Composite End Points to Measure Clinical Events. JAMA, 290, 1456-b-1457.

The correct answer is: Patients consider some component endpoints of the composite measure to be more important than the others

Question 27

Not answered

Marked out of 1

To evaluate the effectiveness of three readily available National Health Service treatments for adolescents (aged 12-18 years) with anorexia nervosa, a multicentre randomised controlled trial of 167 young people comparing in-patient, specialist out-patient and general child and adolescent mental health service (CAMHS) treatment was carried out. There were no significant differences between the samples recruited from two different sites ($n_1=80$ and $n_2=87$). The study period was two years; 17 patients in the CAMHS group, 14 in the outpatient group and 29 in the inpatient group did not receive the intended intervention. Based on categorical outcome assessments, 18% of patients in CAMHS service, 15% in specialist out-patient service and 21% in inpatient service demonstrated a predefined satisfactory clinical response. (Gowers et al., Clinical effectiveness of treatments for anorexia nervosa in adolescents: Randomised controlled trial. The British Journal of Psychiatry 2007; 191: 427-435). The similarity of treatment groups with respect to baseline characteristics most likely to be due to

Select one:

- ☐ Comparable attrition rates
- ☐ Use of multiple centres
- ☐ Use of randomization
- ☐ Use of intention-to-treat analysis
- ☐ Use of blinding

Your answer is incorrect.

The purpose of randomization is to control or balance out the effects of any extraneous variables. Random allocation ensures that any differences between the two study groups at the beginning of the study are due to random variation and not the selection per se. Blinding happens after selection is complete; it cannot nullify baseline differences. The same applies to the intention to treat analysis.

The correct answer is: Use of randomization

Question 28

Not answered

Marked out of 1

To evaluate the effectiveness of three readily available National Health Service treatments for adolescents (aged 12-18 years) with anorexia nervosa, a multicentre randomised controlled trial of 167 young people comparing inpatient, specialist outpatient and general child and adolescent mental health service (CAMHS) treatment was carried out. There were no significant differences between the samples recruited from two different sites ($n_1=80$ and $n_2=87$). The study period was two years; 17 patients in the CAMHS group, 14 in the outpatient group and 29 in the inpatient group did not receive the intended intervention. Based on categorical outcome assessments, 18% of patients in CAMHS service, 15% in specialist out-patient service and 21% in inpatient service demonstrated a predefined satisfactory clinical response. (Gowers et al., Clinical effectiveness of treatments for anorexia nervosa in adolescents: Randomised controlled trial. The British Journal of Psychiatry 2007; 191: 427-435). The CAMHS group received standard care without any specialised input. This group can be correctly described as

Select one:

- ☐ Placebo group
- ☐ Non-inferiority group
- ☐ Treatment as usual group
- ☐ Blind group
- ☐ Open control group

Your answer is incorrect.

Declaration of Helsinki demands that individual patients in a study 'be assured of the best proven diagnostic and therapeutic method' even in the control group (Rothman and Michels, 1994). This statement clearly discards the use of a placebo as control when a 'proven' treatment exists. This is often the case when considering trials of service delivery and psychological interventions. In some cases, TAU may mean comparing active arm with patients on waiting list. Such methods have obvious issues with ensuring randomness and unbiased selection. Rothman, K. J. and Michels, K. B. (1994) The continuing unethical use of placebo controls. New England Journal of Medicine, 331, 394-398

The correct answer is: Treatment as usual group

Question 29

Not answered

Marked out of 1

To evaluate the effectiveness of three readily available National Health Service treatments for adolescents (aged 12-18 years) with anorexia nervosa, a multicentre randomised controlled trial of 167 young people comparing inpatient, specialist outpatient and general child and adolescent mental health service (CAMHS) treatment was carried out. There were no significant differences between the samples recruited from two different sites ($n_1=80$ and $n_2=87$). The study period was two years; 17 patients in CAMHS group, 14 in the outpatient group and 29 in inpatient group did not receive the intended intervention. Based on categorical outcome assessments, 18% of patients in CAMHS service, 15% in specialist out-patient service and 21% in inpatient service demonstrated a predefined satisfactory clinical response. (Gowers et al., Clinical effectiveness of treatments for anorexia nervosa in adolescents: Randomised controlled trial. The British Journal of Psychiatry 2007; 191: 427-435). The risk of not responding clinically among CAMHS patients was

Select one:

- ☐ 0.45
- ☐ 0.18
- ☐ 0.64
- ☐ 0.09
- ☐ 0.82

Your answer is incorrect.

As 18% of CAMHS patients showed response, the 'non-response' rate is $100-18 = 82\%$. The risk of non-response in this group is 0.82

The correct answer is: 0.82

Question 30

Not answered

Marked out of 1

To evaluate the effectiveness of three readily available National Health Service treatments for adolescents (aged 12-18 years) with anorexia nervosa, a multicentre randomised controlled trial of 167 young people comparing inpatient, specialist outpatient and general child and adolescent mental health service (CAMHS) treatment was carried out. There were no significant differences between the samples recruited from two different sites ($n_1=80$ and $n_2=87$). The study period was two years; 17 patients in CAMHS group, 14 in the outpatient group and 29 in inpatient group did not receive the intended intervention. Based on categorical outcome assessments, 18% of patients in CAMHS service, 15% in specialist out-patient service and 21% in inpatient service demonstrated a predefined satisfactory clinical response. (Gowers et al., Clinical effectiveness of treatments for anorexia nervosa in adolescents: Randomised controlled trial. The British Journal of Psychiatry 2007; 191: 427-435). Although more inpatients failed to complete the study compared to outpatients, all patients were analyzed according to the original assignment. This is best described as

Select one:

- ☐ Intention to treat analysis
- ☐ Use of randomisation
- ☐ Economic analysis
- ☐ Treatment as usual
- ☐ Implied consent research

Your answer is incorrect.

RCTs can undermine randomization if they do not include patients who do not receive their assigned treatment in the final analysis. Such analyses will bias the results if the reasons for nonadherence are related to prognosis or side effects. In a number of randomized trials, patients who did not adhere to their assigned drug regimens have fared worse than those who took their medication as instructed, even after taking into account all known prognostic factors and even when their medications were placebos. Omitting those who do not receive assigned treatment undermines the unbiased comparison provided by randomization. We can prevent this bias when they follow the intention-to-treat principle and analyze all patients in the group to which they were randomized. (JAMA User's Guide to Medical Literature: Essentials of evidence-based clinical practice 2nd ed. Pg 96)

The correct answer is: Intention to treat analysis

Question 31

Not answered

Marked out of 1

To evaluate the effectiveness of three readily available National Health Service treatments for adolescents (aged 12-18 years) with anorexia nervosa, a multicentre randomised controlled trial of 167 young people comparing in-patient (n=57), specialist out-patient(n=55) and general child and adolescent mental health service (CAMHS) treatment(n=55) was carried out. There were no significant differences between the samples recruited from two different sites (n at site1=80 and n at site2=87). The study period was two years; 17 patients in CAMHS group, 14 in the outpatient group and 29 in the inpatient group did not complete the intended intervention. Based on categorical outcome assessments, 18% of patients in CAMHS service, 15% in specialist out-patient service and 21% in inpatient service demonstrated a predefined satisfactory clinical response. (Gowers et al, Clinical effectiveness of treatments for anorexia nervosa in adolescents: Randomised controlled trial. The British Journal of Psychiatry 2007; 191: 427-435). The risk ratio of discontinuation among inpatients to the corresponding risk among CAMHS outpatients is

Select one:

- ☐ 0.03
- ☐ 1.17
- ☐ 0.16666667
- ☐ 1.64
- ☐ 0.857142857

Your answer is incorrect.

The noncompletion rate is 29 out of 57 inpatients; it is 17 out of 55 CAMHS patients. The risk is 0.508 for inpatients and 0.309 for CAMHS. The risk ratio is $0.508/0.309 = 1.64$.

The correct answer is: 1.64

Question 32

Not answered

Marked out of 1

To evaluate the effectiveness of three readily available National Health Service treatments for adolescents (aged 12-18 years) with anorexia nervosa, a multicentre randomised controlled trial of 167 young people comparing in-patient, specialist out-patient and general child and adolescent mental health service (CAMHS) treatment was carried out. There were no significant differences between the samples recruited from two different sites ($n_1=80$ and $n_2=87$). The study period was two years; 17 patients in CAMHS group, 14 in the outpatient group and 29 in the inpatient group did not receive the intended intervention. Based upon categorical outcome assessments, 18% of patients in CAMHS service, 15% in specialist out-patient service and 21% in inpatient service demonstrated a predefined satisfactory clinical response. Intention to treat analysis did not demonstrate statistically significant differences in clinical response. (Gowers et al., Clinical effectiveness of treatments for anorexia nervosa in adolescents: Randomised controlled trial. The British Journal of Psychiatry 2007; 191: 427-435). The results suggest that compared with CAMHS outpatients, those treated with specialised inpatient services have

Select one:

- ☐ Worse clinical response, but fewer drop-outs
- ☐ Better clinical response and fewer drop-outs
- ☐ Better clinical response, but more drop-outs
- ☐ Worse clinical response but no dropouts
- ☐ Equivalent clinical response, but more drop-outs

Your answer is incorrect.

As there is no demonstrable statistical difference between the groups, we can conclude that the response was equivalent. But in the absence of any mention of the statistical significance of dropout rates, we can still see a perceivable difference among three groups with a good possibility that drop-outs are lesser with outpatient services.

The correct answer is: Equivalent clinical response, but more drop-outs

Question **33**

Not answered

Marked out of 1

A trial comparing an intervention and control concluded that the absolute risk reduction for the primary outcome is zero ($ARR = 0$). Which of the following is the accurate interpretation

Select one:

- ☐ There is no difference between the 2 groups
- ☐ The baseline risk of the outcome is zero
- ☐ The treatment group did not have any primary outcome event on the record.
- ☐ The control group did not have any primary outcome event on the record.
- ☐ The sample studied did not experience even one primary outcome event

Your answer is incorrect.

$ARR = 0$ means that there is no difference between the two groups (thus, the treatment had no effect)

The correct answer is: There is no difference between the 2 groups



Question **34**

Not answered

Marked out of 1

In a trial comparing antipsychotics, the relapse rate in the group taking chlorpromazine is 45%. The relapse rate in the group taking the evaluated new drug is 50%. The absolute reduction in the risk of relapse on prescribing the new drug will be

Select one:

- ☐ 0.5
- ☐ 0.05
- ☐ 1
- ☐ 50
- ☐ 5

Your answer is incorrect.

ARR = control event rate - experimental event rate. $50 - 45\% = 5\% = 0.05$. Note that ARR is generally an absolute value of the difference between the outcome rates in the two treatment groups; so one need not worry about the formula CER - EER or EER - CER as long as one understands the significance of the event described.

The correct answer is: 0.05

Question 35

Not answered

Marked out of 1

In a study comparing drug A and placebo control, 20 out of 200 patients taking drug A die after three years. 25 out of 225 patients taking the placebo die after three years. If death is the outcome of interest, the relative risk of death in patients taking drug A compared to placebo is

Select one:

- ☐ 25
- ☐ 25/225
- ☐ 5/225
- ☐ 0.9
- ☐ 40066

Your answer is incorrect.

RR is EER/CER. Here the event rate in the experimental group is 20/200. The same in control (placebo) group is 25/225. The RR - $\frac{1}{10} \times \frac{1}{9} = \frac{9}{10} = 0.9$

The correct answer is: 0.9



Question 36

Not answered

Marked out of 1

In a study comparing drug A and placebo control, 20 out of 200 patients taking drug A die after three years. 25 out of 225 patients taking the placebo die after three years. If death is the outcome of interest, the amount by which death is reduced by administering drug A for three years compared to placebo is

Select one:

- ☐ 0.2
- ☐ 0.05
- ☐ 0.9
- ☐ 0.25
- ☐ 0.1

Your answer is incorrect.

RRR (relative risk reduction) describes the reduction in events (e.g. 10% fewer deaths) while ARR describes the reduction in persons affected by the event (e.g. 1 in 90 was saved). ARR gives us a concrete number to determine the size of the effect. One advantage of the ARR is that it exemplifies the relationship between the therapeutic effort (treating 90 people) and the clinical yield (saving one life). This is made clearer by calculating NNT using ARR. Here RRR is given by $1 - RR = 1 - 0.9 = 0.1$ or 10%

The correct answer is: 0.1

Question 37

Not answered

Marked out of 1

A researcher administers the active medication for a certain period, before carrying out randomization for an RCT designed to evaluate a drug's efficacy. Which of the following best describes the purpose of performing the above run-in phase before randomising eligible participants

Select one:

- ☐ To obtain informed consent
- ☐ To enable intention to treat analysis
- ☐ To conceal allocation
- ☐ To improve compliance
- ☐ To identify placebo effect

Your answer is incorrect.

Many RCTs are preceded by a "placebo run-in," in which all patients are given a placebo. The belief is that this will weed out non-compliers before randomisation, eliminate placebo responders, ensure that patients are stable and wash out previous treatment, or simply to provide a period for a baseline measurement. In a similar vein, trials could also employ run-in of active medications to detect the degree of compliance.

The correct answer is: To improve compliance

Question 38

Not answered

Marked out of 1

Mood stabiliser Evaluation Programme (MEP) is a multicentre randomised control trial to evaluate the effectiveness of lithium, valproate and quetiapine for relapse prevention in bipolar disorder. The 18 months follow-up showed a relapse rate of 68% in patients who were less compliant with lithium compared to 15% in patients who adhered to lithium (having therapeutic levels of lithium 90% of times when blood samples are obtained). The result was shown to be significant with $p < 0.003$. Which of the following is correct

Select one:

- ☐ Lithium was very beneficial to the patients who took it reliably
- ☐ Either lithium or the factors related to treatment-adherence is/are associated with reduced relapse rates
- ☐ Lithium is not effective unless at least 90% of the plasma monitoring return therapeutic levels.
- ☐ While evaluating effectiveness the data from treatment adherent patients only must be used
- ☐ Lack of placebo control invalidates this study

Your answer is incorrect.

The independent variable (lithium) is affected by another variable - compliance. There may be many unseen variables associated with higher compliance that may tilt the balance in favour of compliers - e.g. Strong social network, good degree of insight, higher educational levels, therapeutic relationship, availability of contact with services etc. Hence the conclusion 'Either lithium or the factors related to treatment adherence is/are associated with reduced relapse rates' is the most appropriate.

The correct answer is: Either lithium or the factors related to treatment-adherence is/are associated with reduced relapse rates

Question 39

Not answered

Marked out of 1

Which of the following is probably the most commonly reported measures of treatment effect in efficacy trials (RCTs)

Select one:

- ☐ Control event rate
- ☐ Proportion treated
- ☐ Relative risk reduction
- ☐ Odds ratio
- ☐ Experimental event rate

Your answer is incorrect.

The most commonly reported measure of dichotomous treatment effects is the complement of the RR, the RRR. It is expressed as a percentage: $1 - (EER/CER)$ Guyatt G Rennie D et al. Users' guide to the medical literature: JAMA evidence 2nd ed. McGrawHill 2008. Pg 76

The correct answer is: Relative risk reduction

Question 40

Not answered

Marked out of 1

A randomised controlled trial was conducted in Norway to determine whether a systematic approach to the treatment of pain can reduce agitation in people with moderate to severe dementia living in nursing homes. 352 residents with moderate to severe dementia and clinically significant behavioural disturbances were recruited from 60 clusters (single independent nursing home units) in 18 nursing homes within five municipalities of western Norway. The nursing home units were randomised to use a stepwise protocol for the treatment of pain in eligible subjects for eight weeks with additional follow-up four weeks after the end of treatment (33 clusters; n=175) or to usual treatment (control, 27 clusters; n=177). Sample size was calculated on the basis of clinically meaningful differences in agitation scores after taking into account the expected intracluster correlation and dropout rates. Participants in the intervention group received individual daily treatment of pain for eight weeks according to the stepwise protocol, with paracetamol (acetaminophen), morphine, buprenorphine transdermal patch, or pregabalin. The control group received usual treatment and care. The primary outcome measure was agitation scores on Cohen-Mansfield agitation inventory. Agitation was significantly reduced in the intervention group compared with control group after eight weeks (results shown in the below table): the average reduction in scores for agitation was 17% (treatment effect estimate -7.0, 95% confidence interval -3.7 to -10.3). Treatment of pain was also significantly beneficial for the overall severity of neuropsychiatric symptoms (-9.0, -5.5 to -12.6) and pain (-1.3, -0.8 to -1.7), but the groups did not differ significantly for activities of daily living or cognition. In the intervention group, there was a worsening of scores on the Cohen-Mansfield agitation inventory between week 8 and week 12, after withdrawal of pain treatment. The authors concluded that a systematic approach to the management of pain could alleviate agitation and could reduce the number of unnecessary prescriptions for psychotropic drugs in residents of nursing homes with moderate to severe dementia. What kind of sampling has been used?

Table 3 | Comparison of Cohen-Mansfield agitation inventory (CMAI) total score between control and intervention (stepwise protocol for treatment of pain) groups using repeated measures analysis of covariance (ANCOVA)*

Week	Mean (SD) CMAI total score		Effect of intervention on CMAI total†		Intraclass correlation coefficient‡
	Control group	Intervention group	Estimate (95% CI)	P value	
0	56.2 (16.1), n=177	56.5 (15.2), n=175	—	—	0.162
2	53.9 (17.0), n=161	52.0 (19.5), n=158	-3.6 (-0.5 to -6.7)	0.022	0.261
4	52.5 (16.3), n=160	49.4 (19.0), n=148	-4.1 (-0.9 to -7.4)	0.012	0.231
8	52.8 (16.8), n=157	46.9 (18.7), n=147	-7.0 (-3.7 to -10.3)	<0.001	0.226
12	52.5 (16.0), n=152	50.3 (20.3), n=142	-3.2 (0.1 to -6.4)	0.058	0.253

*Baseline score as covariate and least squares weighted by number of patients within cluster; P value from multivariate test of intervention was 0.002, and cross effect between week and intervention was <0.001.

†Variable estimate by week of effect of intervention on CMAI score from estimated model.

‡Proportion of total variance between clusters, and measured within framework of ANCOVA.

Select one:

- ☐ Simple random sampling
- ☐ Effective sampling
- ☐ Cluster sampling
- ☐ Quota sampling
- ☐ Multistage sampling

Your answer is incorrect.

This study has used a cluster sampling approach for randomisation.

The correct answer is: Cluster sampling

Question **41**

Not answered

Marked out of 1

A randomised controlled trial was conducted in Norway to determine whether a systematic approach to the treatment of pain can reduce agitation in people with moderate to severe dementia living in nursing homes. 352 residents with moderate to severe dementia and clinically significant behavioural disturbances were recruited from 60 clusters (single independent nursing home units) in 18 nursing homes within five municipalities of western Norway. The nursing home units were randomised to use a stepwise protocol for the treatment of pain in eligible subjects for eight weeks with additional follow-up four weeks after the end of treatment (33 clusters; n=175) or to usual treatment (control, 27 clusters; n=177). Sample size was calculated on the basis of clinically meaningful differences in agitation scores after taking into account the expected intracluster correlation and dropout rates. Participants in the intervention group received individual daily treatment of pain for eight weeks according to the stepwise protocol, with paracetamol (acetaminophen), morphine, buprenorphine transdermal patch, or pregabalin. The control group received usual treatment and care. The primary outcome measure was agitation scores on Cohen-Mansfield agitation inventory. Agitation was significantly reduced in the intervention group compared with control group after eight weeks (results shown in the below table): the average reduction in scores for agitation was 17% (treatment effect estimate -7.0, 95% confidence interval -3.7 to -10.3). Treatment of pain was also significantly beneficial for the overall severity of neuropsychiatric symptoms (-9.0, -5.5 to -12.6) and pain (-1.3, -0.8 to -1.7), but the groups did not differ significantly for activities of daily living or cognition. In the intervention group, there was a worsening of scores on the Cohen-Mansfield agitation inventory between week 8 and week 12, after withdrawal of pain treatment. The authors concluded that a systematic approach to the management of pain could alleviate agitation and could reduce the number of unnecessary prescriptions for psychotropic drugs in residents of nursing homes with moderate to severe dementia. With respect to the power of this study, which of the following is true?

Table 3 | Comparison of Cohen-Mansfield agitation inventory (CMAI) total score between control and intervention (stepwise protocol for treatment of pain) groups using repeated measures analysis of covariance (ANCOVA)*

Week	Mean (SD) CMAI total score		Effect of intervention on CMAI total†		Intracluster correlation coefficient‡
	Control group	Intervention group	Estimate (95% CI)	P value	
0	56.2 (16.1), n=177	56.5 (15.2), n=175	—	—	0.162
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*Baseline score as covariate and least squares weighted by number of patients within cluster; P value from multivariate test of intervention was 0.002, and cross effect between week and intervention was <0.001.

†Variable estimate by week of effect of intervention on CMAI score from estimated model.

‡Proportion of total variance between clusters, and measured within framework of ANCOVA.

Select one:

- ☐ This study does not have the ability to detect the true difference between the intervention group and control group
- ☐ Higher power can be achieved using same number of subjects but without cluster -sampling
- ☐ Cluster sampling has reduced the need for recruiting more subjects
- ☐ Cluster sampling has not affected the power calculations of this trial
- ☐ Higher power can be achieved using same number of subjects but shorter period of follow-up

Your answer is incorrect.

By using cluster-sampling the power of a study is reduced, thus more subjects (larger sample size) are required in order to have the ability to observe a difference.

The correct answer is: Higher power can be achieved using same number of subjects but without cluster -sampling



Question 42

Not answered

Marked out of 1

A randomised controlled trial was conducted in Norway to determine whether a systematic approach to the treatment of pain can reduce agitation in people with moderate to severe dementia living in nursing homes. 352 residents with moderate to severe dementia and clinically significant behavioural disturbances were recruited from 60 clusters (single independent nursing home units) in 18 nursing homes within five municipalities of western Norway. The nursing home units were randomised to use a stepwise protocol for the treatment of pain in eligible subjects for eight weeks with additional follow-up four weeks after the end of treatment (33 clusters; n=175) or to usual treatment (control, 27 clusters; n=177). Sample size was calculated on the basis of clinically meaningful differences in agitation scores after taking into account the expected intracluster correlation and dropout rates. Participants in the intervention group received individual daily treatment of pain for eight weeks according to the stepwise protocol, with paracetamol (acetaminophen), morphine, buprenorphine transdermal patch, or pregabalin. The control group received usual treatment and care. The primary outcome measure was agitation scores on Cohen-Mansfield agitation inventory. Agitation was significantly reduced in the intervention group compared with control group after eight weeks (results shown in the below table): the average reduction in scores for agitation was 17% (treatment effect estimate -7.0, 95% confidence interval -3.7 to -10.3). Treatment of pain was also significantly beneficial for the overall severity of neuropsychiatric symptoms (-9.0, -5.5 to -12.6) and pain (-1.3, -0.8 to -1.7), but the groups did not differ significantly for activities of daily living or cognition. In the intervention group, there was a worsening of scores on the Cohen-Mansfield agitation inventory between week 8 and week 12, after withdrawal of pain treatment. The authors concluded that a systematic approach to the management of pain could alleviate agitation and could reduce the number of unnecessary prescriptions for psychotropic drugs in residents of nursing homes with moderate to severe dementia. In this trial, the term intracluster correlation refers to the proportion of the total variability in the data that is due which one of the following?

Table 3| Comparison of Cohen-Mansfield agitation inventory (CMAI) total score between control and intervention (stepwise protocol for treatment of pain) groups using repeated measures analysis of covariance (ANCOVA)*

Week	Mean (SD) CMAI total score		Effect of intervention on CMAI total†		Intraclass correlation coefficient‡
	Control group	Intervention group	Estimate (95% CI)	P value	
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*Baseline score as covariate and least squares weighted by number of patients within cluster; P value from multivariate test of intervention was 0.002, and cross effect between week and intervention was <0.001.

†Variable estimate by week of effect of intervention on CMAI score from estimated model.

‡Proportion of total variance between clusters, and measured within framework of ANCOVA.

Select one:

- ☐ Variability induced by different stepwise treatment options
- ☐ Variability expected among patients in the same nursing home unit
- ☐ Variability expected between the municipalities
- ☐ Variability expected between the nursing home units
- ☐ Variability expected between patients receiving active and control interventions

Your answer is incorrect.

Effective sample size (ESS) is used in place of actual sample size when undertaking power calculations in a cluster randomised trial. Sample size, in general, refers to the number of units of independent participants recruited for a trial. In a cluster RCT, individuals are not recruited but 'clusters' are recruited. So the effective sample size is much smaller than the actual number of individual participants. To calculate ESS, statisticians use a measure called 'variance inflation factor' or 'design effect' which is derived from an intraclass coefficient. (This concept is

very similar to the notion of intraclass correlation coefficient used when studying inter-rater reliability). A high intracluster coefficient will reduce the effective sample size. Intracluster correlation ρ refers to the proportion of total variability of the outcome measure (sum of between-cluster and within-cluster variances) attributed to the between-cluster variability. It can vary between 0 and 1. A very high value for ρ (close to 1) is bad news, as it suggests that all of the variability is due to between-cluster differences, and the responses within a cluster are highly correlated. In other words, this means that the entire cluster is behaving as a single observational unit (equivalent to just one patient), thus reducing the effective sample size to a value closer to the total number of clusters recruited. The calculation of ρ usually requires a pilot study.

The correct answer is: Variability expected between the nursing home units



Question 43

Not answered

Marked out of 1

A randomised controlled trial was conducted in Norway to determine whether a systematic approach to the treatment of pain can reduce agitation in people with moderate to severe dementia living in nursing homes. 352 residents with moderate to severe dementia and clinically significant behavioural disturbances were recruited from 60 clusters (single independent nursing home units) in 18 nursing homes within five municipalities of western Norway. The nursing home units were randomised to use a stepwise protocol for the treatment of pain in eligible subjects for eight weeks with additional follow-up four weeks after the end of treatment (33 clusters; n=175) or to usual treatment (control, 27 clusters; n=177). Sample size was calculated on the basis of clinically meaningful differences in agitation scores after taking into account the expected intracluster correlation and dropout rates. Participants in the intervention group received individual daily treatment of pain for eight weeks according to the stepwise protocol, with paracetamol (acetaminophen), morphine, buprenorphine transdermal patch, or pregabalin. The control group received usual treatment and care. The primary outcome measure was agitation scores on Cohen-Mansfield agitation inventory. Agitation was significantly reduced in the intervention group compared with control group after eight weeks (results shown in the below table): the average reduction in scores for agitation was 17% (treatment effect estimate -7.0, 95% confidence interval -3.7 to -10.3). Treatment of pain was also significantly beneficial for the overall severity of neuropsychiatric symptoms (-9.0, -5.5 to -12.6) and pain (-1.3, -0.8 to -1.7), but the groups did not differ significantly for activities of daily living or cognition. In the intervention group, there was a worsening of scores on the Cohen-Mansfield agitation inventory between week 8 and week 12, after withdrawal of pain treatment. The authors concluded that a systematic approach to the management of pain could alleviate agitation and could reduce the number of unnecessary prescriptions for psychotropic drugs in residents of nursing homes with moderate to severe dementia. Considering the results reported in this study, which of the following can be considered a limitation?

Table 3| Comparison of Cohen-Mansfield agitation inventory (CMAI) total score between control and intervention (stepwise protocol for treatment of pain) groups using repeated measures analysis of covariance (ANCOVA)*

Week	Mean (SD) CMAI total score		Effect of intervention on CMAI total†		Intracluster correlation coefficient‡
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*Baseline score as covariate and least squares weighted by number of patients within cluster; P value from multivariate test of intervention was 0.002, and cross effect between week and intervention was <0.001.

†Variable estimate by week of effect of intervention on CMAI score from estimated model.

‡Proportion of total variance between clusters, and measured within framework of ANCOVA.

Select one:

- ☐ Worsening of agitation by week 12 indicating lack of sustained treatment effects
- ☐ The appearance of beneficial effect as early as week 2
- ☐ Poor generalizability of results to other dementia sufferers living in nursing homes
- ☐ Lack of improvement in cognition and activities of daily living
- ☐ Improvement in agitation occurred in both intervention and control clusters across time

Your answer is incorrect.

Poor generalizability is generally a worry with most RCTs. In this study this has been addressed to some extent by following several naturalistic approaches e.g. using stepped care pain management model rather than a fixed dose/drug for analgesia, recruiting from several nursing homes instead of selecting a single unit, comparison with treatment as usual rather than a sham or placebo etc. Worsening of agitation occurred over the four weeks of follow-up after the intervention had stopped - not when the treatment was continued. This is not a

limitation. In fact, this suggests that the treatment has had a positive effect that ceased upon stopping the treatment. Analgesic treatments are expected to work within days to relieve pain but are not expected to reverse dementia; so the lack of improvement in cognition and ADLs is irrelevant. One limitation is the observation that notable drop in agitation occurred in the control group as well (weeks 2 and 4). This increases the concern for non-specific effects e.g. social interaction and staff-training, etc.

The correct answer is: Improvement in agitation occurred in both intervention and control clusters across time



Question **44**

Not answered

Marked out of 1

A randomised controlled trial was conducted in Norway to determine whether a systematic approach to the treatment of pain can reduce agitation in people with moderate to severe dementia living in nursing homes. 352 residents with moderate to severe dementia and clinically significant behavioural disturbances were recruited from 60 clusters (single independent nursing home units) in 18 nursing homes within five municipalities of western Norway. The nursing home units were randomised to use a stepwise protocol for the treatment of pain in eligible subjects for eight weeks with additional follow-up four weeks after the end of treatment (33 clusters; n=175) or to usual treatment (control, 27 clusters; n=177). Sample size was calculated on the basis of clinically meaningful differences in agitation scores after taking into account the expected intracluster correlation and dropout rates. Participants in the intervention group received individual daily treatment of pain for eight weeks according to the stepwise protocol, with paracetamol (acetaminophen), morphine, buprenorphine transdermal patch, or pregabalin. The control group received usual treatment and care. The primary outcome measure was agitation scores on Cohen-Mansfield agitation inventory. Agitation was significantly reduced in the intervention group compared with control group after eight weeks (results shown in the below table): the average reduction in scores for agitation was 17% (treatment effect estimate -7.0, 95% confidence interval -3.7 to -10.3). Treatment of pain was also significantly beneficial for the overall severity of neuropsychiatric symptoms (-9.0, -5.5 to -12.6) and pain (-1.3, -0.8 to -1.7), but the groups did not differ significantly for activities of daily living or cognition. In the intervention group, there was a worsening of scores on the Cohen-Mansfield agitation inventory between week 8 and week 12, after withdrawal of pain treatment. The authors concluded that a systematic approach to the management of pain could alleviate agitation and could reduce the number of unnecessary prescriptions for psychotropic drugs in residents of nursing homes with moderate to severe dementia. Which of the following is true regarding study?

Table 3 | Comparison of Cohen-Mansfield agitation inventory (CMAI) total score between control and intervention (stepwise protocol for treatment of pain) groups using repeated measures analysis of covariance (ANCOVA)*

Week	Mean (SD) CMAI total score		Effect of intervention on CMAI total†		Intraclass correlation coefficient‡
	Control group	Intervention group	Estimate (95% CI)	P value	
0	56.2 (16.1), n=177	56.5 (15.2), n=175	—	—	0.162
2	53.9 (17.0), n=161	52.0 (19.5), n=158	-3.6 (-0.5 to -6.7)	0.022	0.261
4	52.5 (16.3), n=160	49.4 (19.0), n=148	-4.1 (-0.9 to -7.4)	0.012	0.231
8	52.8 (16.8), n=157	46.9 (18.7), n=147	-7.0 (-3.7 to -10.3)	<0.001	0.226
12	52.5 (16.0), n=152	50.3 (20.3), n=142	-3.2 (0.1 to -6.4)	0.058	0.253

*Baseline score as covariate and least squares weighted by number of patients within cluster; P value from multivariate test of intervention was 0.002, and cross effect between week and intervention was <0.001.

†Variable estimate by week of effect of intervention on CMAI score from estimated model.

‡Proportion of total variance between clusters, and measured within framework of ANCOVA.

Select one:

- ☐ Baseline severity of agitation has confounded the reported results
- ☐ All subjects in the intervention group responded positively to the treatment
- ☐ The control group showed greater reduction in agitation by week 12
- ☐ The intervention group had higher agitation at the baseline
- ☐ Significant group differences are noted in weeks 2, 4 and 8

Your answer is incorrect.

The results presented in the table refer to average scores obtained from each group across the five time points. It is possible that in the intervention group, some patients improved while others showed either no improvement or worsened. This cannot be deducted from the results shown in this table. From the p-values displayed, it is clear that significant group differences were present in weeks 2, 4 and 8 (with

progressive improvement in the intervention group) but not at week 12. ANCOVA test has been used with baseline severity as a covariate; so it is unlikely that the reported results are confounded by the baseline.

The correct answer is: Significant group differences are noted in weeks 2, 4 and 8

Question **45**

Not answered

Marked out of 1

Which of the following is the least adequate method of allocation in a clinical trial?

Select one:

- ☐ Odd/even last digit of date of birth
- ☐ Simple randomisation with a computer
- ☐ Permuted block randomisation
- ☐ Minimisation
- ☐ Coin Toss

Your answer is incorrect.

The 'date-of-birth' method is, in fact, a quasi-randomisation technique.

The correct answer is: Odd/even last digit of date of birth

Question 46

Not answered

Marked out of 1

Which of the following types of RCTs will require the highest sample size assuming all other determinants of sample size are constant?

Select one:

- ☐ Equivalence trials
- ☐ Superiority trials with large effect size
- ☐ One tailed superiority trials
- ☐ Cross over superiority trials
- ☐ Non inferiority trials with large margin of difference

Your answer is incorrect.

. Equivalence trials by definition will have zero effect size difference to be demonstrated. Theoretically infinite sample size will be required to conduct a pure point equivalence trial. In practical terms, only ranges of effects are compared for overlap between two groups within a predefined value to be termed as equivalent.

The correct answer is: Equivalence trials

Question 47

Not answered

Marked out of 1

Which one of the following is a unique advantage of RCTs compared to other methods of research?

Select one:

- ☐ Allows prospective study design
- ☐ Allows systematic review of results
- ☐ Potentially eradicates bias by comparing two almost identical groups
- ☐ None of the above
- ☐ Produces generalisable results

Your answer is incorrect.

Systematic review of results is possible even with cohort or case control studies. Both cohort and RCT designs allow prospective observations. One often quoted disadvantage of RCTs is lack of generalisability of results from a highly controlled experiment to somewhat heterogeneous and changeable clinical practice. Randomisation potentially reduces bias that may occur during allocation or selection; controlling intervention and blinding reduce bias that may occur when intervention is given or outcome is measured.

The correct answer is: Potentially eradicates bias by comparing two almost identical groups

Question 48

Not answered

Marked out of 1

You have identified a clinical trial comparing two different antidepressant drugs. Which of the following measures will best assist you when discussing the clinical effect of these two drugs with a patient?

Select one:

- ☐ ARP
- ☐ RRR
- ☐ ARR
- ☐ OR
- ☐ NNT

Your answer is incorrect.

NNT stands for the number needed to treat. It is the most often used clinical index to explain the efficacy of an intervention to individual patients.

The correct answer is: NNT



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